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Commentary

ChatGPT for mechanobiology and medicine: A perspective

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ABSTRACT

ChatGPT has garnered significant attention for its impressive capabilities across various domains, including medicine and mechanobiology. In order to facilitate the integration of ChatGPT into research, this paper explores the applications of ChatGPT in these domains, focusing on its usage in (1) reading and writing, (2) retrieval and knowledge management, and (3) computation, simulation, and visualization. Meanwhile, this study acknowledges the limitations and challenges associated with ChatGPT's usage. We investigate the interaction between ChatGPT and external tools in these applications and advocate for the integration of more powerful tools in these research areas into ChatGPT to further expand its potential applications in medicine and mechanobiology.

ChatGPT (Chat Generative Pre-trained Transformer) is an artificial intelligence conversational agent built on the state-of-the-art transformer-based LLM (Large Language Model) GPT released by OpenAI in November 2022. With proper prompts, this versatile chatbot exhibits fluent and sophisticated natural language processing abilities. Recently, OpenAI upgraded the underlying architecture of ChatGPT from GPT-3.5 to GPT-4, enabling it to handle more complex multimodal prompts, including text and images.

Due to its exceptional performance and promising potential across various dialogue scenarios, ChatGPT has rapidly gained prominence within a short period. Researchers from diverse domains have made vigorous efforts to exploit the capabilities of ChatGPT. In the field of medicine, ChatGPT has demonstrated a passing score in the United States Medical Licensing Examination [1]. Moreover, within specialized domains like clinical toxicology, ChatGPT has exhibited competence in providing comprehensive answers to questions related to typical acute organophosphate poisoning cases [2]. These studies showcase the extensive expertise of ChatGPT and highlight its remarkable potential to drive transformative changes in the medical industry.

Despite the extraordinary capabilities of ChatGPT, it is crucial to acknowledge its limitations. Generative LLMs, including GPT, often exhibit a phenomenon called "hallucination," which refers to the generation of nonsensical or unfaithful content that deviates from the provided source [3]. Consequently, the outputs of ChatGPT are not always reliable due to the possibility of hallucinations. Additionally, ChatGPT occasionally makes simple arithmetic errors that a calculator would typically avoid, restricting its usability in computational tasks. Furthermore, data privacy and fairness pose significant concerns when ChatGPT is employed for personal use.

Considering the aforementioned benefits and drawbacks, we investigate the current and potential impacts of ChatGPT in the fields of medicine and mechanobiology. In addition, to explore its research and clinical capabilities, we evaluate the reliability and limitations of ChatGPT in specific application scenarios. We further particularize its interactions with other practical tools to expand its application foreground, as Fig. 1 shows.

1. Reading and writing

ChatGPT has demonstrated impressive capabilities in context-based tasks within the medical field. Researchers have already utilized ChatGPT for a range of purposes, including but not limited to, summarizing academic articles and scientific reports, describing trends in charts and tables, explaining images using GPT-4, fluent translation of texts from non-native languages, grammar correction during writing, and generating figures, tables, and corresponding LaTeX code. With sufficient context, ChatGPT functions as both a reliable reader and writer.

However, one of the major challenges faced by ChatGPT in reading and writing tasks is domain adaptation. ChatGPT is initially designed for general usage, and although it has surprisingly showcased its abilities in zero-shot settings, studies have reported instances of missing key findings in tasks such as medical paper abstract generation [4] and simplifying radiology reports [5]. Fine-tuning the model using medical texts can enhance its understanding of domain-specific terminology and expressions. However, the parameters of ChatGPT are inaccessible, which means the ChatGPT model cannot be fine-tuned like other open-source LLMs. Furthermore, performance under a few-shot setting, where users

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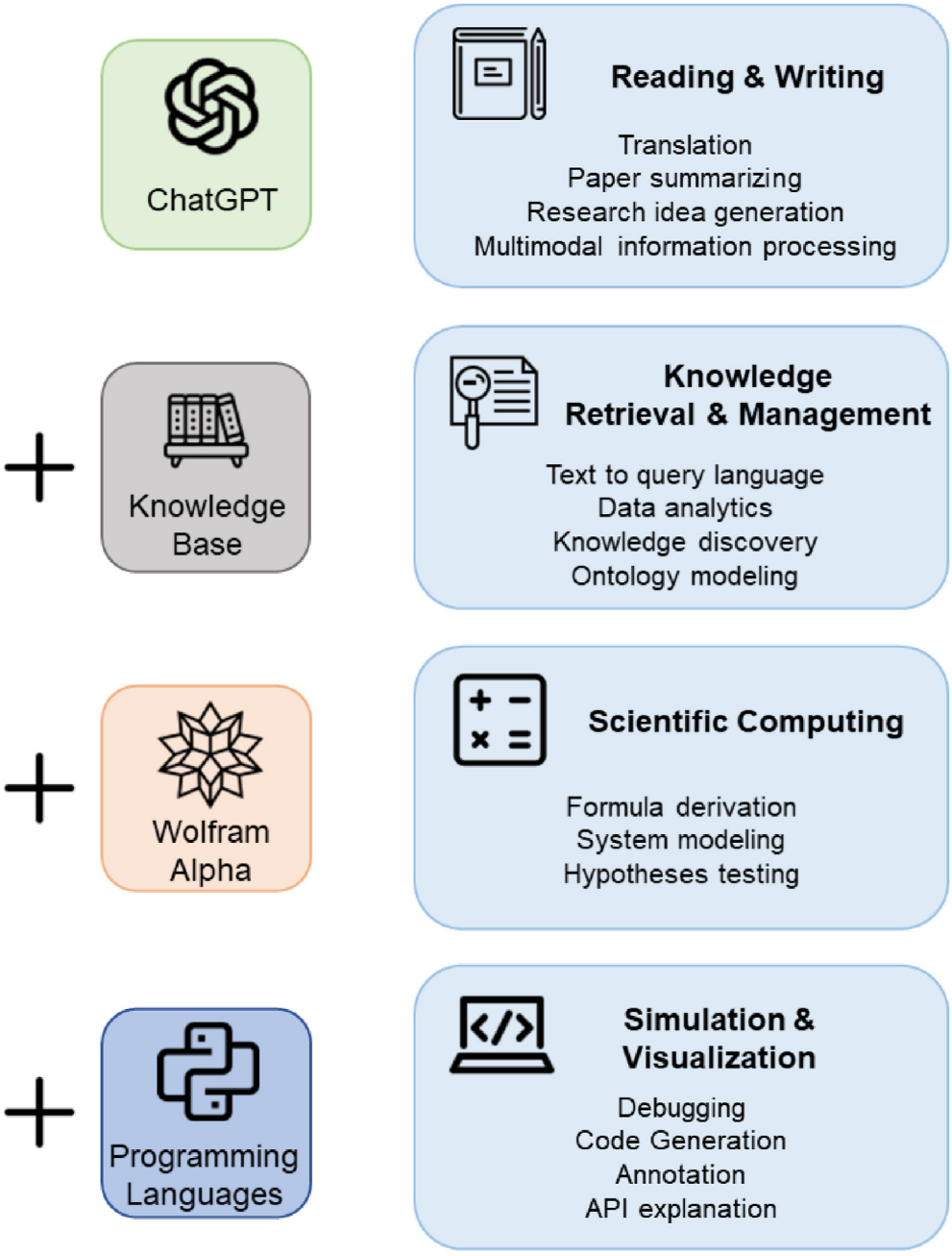


Fig. 1. The current and potential application of ChatGPT for Mechanobiology and Medicine. The capability of ChatGPT can be enhanced by powerful tools to provide more dependable and intelligent assistance.

provide even very few examples, is typically superior to the zero-shot setting on correctness and completeness [6]. This finding aligns with the fact that ChatGPT is unsuitable for conducting original research or offering advice based on limited information. Moreover, the ethical concerns surrounding the usage of ChatGPT in scientific writing remain the subject of ongoing debate.

2. Knowledge retrieval and management

ChatGPT was trained on a vast corpus from a wide range of internet-available materials, including books, articles, websites, and other texts. This extensive training data enables ChatGPT to possess a broad understanding of human language. When prompted with questions, ChatGPT confidently responds based on its training data. However, research indicates that the errors are still present in over one-third of the samples [7,8].

Several factors contribute to these errors. Firstly, ChatGPT lacks up-to-date knowledge as its training process concluded in September 2021. Furthermore, the knowledge learned by ChatGPT is implicitly represented within the parameters of the GPT model rather than in explicit contexts or structural forms, making knowledge retrieval from the model itself more challenging. Additionally, the absence of related material can mislead the model's responses.

Integrating external documents and knowledge into ChatGPT proves to be a practical solution. ChatGPT benefits from accessing additional information beyond its pre-training data. By incorporating relevant external documents, ChatGPT can enhance its responses and reduce the occurrence of errors. Experimental results in passage re-ranking demonstrate the effectiveness of this approach, as ChatGPT shows the ability to handle queries involving documents by utilizing well-designed instructions and employing a sliding windows strategy [9]. Furthermore,

when dealing with structured data stored in databases, ChatGPT can assist users in performing complex database queries by translating natural language into query languages such as SQL and SPARQL. This integration of external knowledge and document retrieval capabilities empowers ChatGPT to serve as a reliable assistant, saving time and effort during interactions with databases.

In this application scenario, ChatGPT serves as a reliable assistant. By referring directly to the provided documents, the issue of hallucination in ChatGPT's responses can be mitigated. Nevertheless, human verification remains necessary to avoid errors.

3. Computation, simulation, and visualization

Computation, simulation, and visualization are vital research methods in medical and mechanobiology studies. These methods enable researchers to model complex systems, test hypotheses, predict outcomes, analyze data, and explore phenomena that are otherwise inaccessible. By employing these techniques, our understanding of biological processes is advanced, leading to improvements in medical applications and interventions.

Regarding computational tasks, an evaluation of ChatGPT conducted on arithmetic tasks indicates that ChatGPT achieves impressive performance on most math quizzes [10]. However, since ChatGPT does not have a built-in calculator module, it is prone to arithmetic mistakes. To overcome this limitation, OpenAI recently enhanced ChatGPT with the Wolfram Alpha plugin, enabling access to accurate and reliable computational capabilities and knowledge. This revolutionary upgrade eliminates the aforementioned drawback. With the assistance of Wolfram Alpha, ChatGPT can now solve differential equations, inspiring and accelerating future mathematical research in mechanobiology.

The codex model, trained alongside the GPT model, is another essential component of ChatGPT. When provided with programming requirements as prompts, codex empowers ChatGPT to generate complete code or useful code snippets. Researchers can leverage this capability to develop simulation tools by translating physical forces in biological phenomena into mathematical forms, such as differential equations [11] or cellular automata [12]. For instance, by prompting ChatGPT with "Generate a cellular automata simulation program with Python," it can provide a code for simulating the "Game of Life." Programmers can significantly benefit from ChatGPT by interactively raising appropriate prompts, such as requesting explanations of API usage, annotating complex code snippets, or even assisting in debugging.

Besides simulation, ChatGPT can also contribute to improving the efficiency and quality of visualization results. It is capable of generating various components of visualizations, including HTML files, JavaScript, and CSS. Even users with limited coding skills can utilize ChatGPT to complete the entire visualization process.

The pitfalls and limitations of ChatGPT-assisted programming are negligible compared with its convenience. Occasionally, ChatGPT provides codes with bugs or errors. So the engineers are supposed to give feedback from the code compiler or interpreter to ChatGPT and ask it to improve its code.

4. Conclusion

Overall, it can be observed that ChatGPT's capability is sufficient to accelerate research in mechanobiology and medicine. Its ability to handle contexts makes it a valuable tool for reading and writing in research progress. By incorporating external documents and knowledge, ChatGPT bridges the gap between users and information, facilitating interactive information retrieval. Furthermore, ChatGPT provides excellent code assistance, enabling individuals with limited coding skills to develop computation, simulation, and visualization tools. We believe exploring more interaction with other tools is a promising way to enhance the capability of ChatGPT. We also advocate building such an ecosystem based on ChatGPT and powerful research tools.

Despite ChatGPT's great potential in various application scenarios, its drawbacks and limitations should not be neglected. Hallucination is yet an obstacle for future LLM applications in mechanobiology and medicine, and human verification is inevitable for correctness, security, and privacy.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

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